

Thanadol (Pleng) Chomphoochan

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EDUCATION

Massachusetts Institute of Technology

Cambridge, MA

M.Eng. in Electrical Engineering and Computer Science · GPA: 5.0/5.0

May 2025

B.Sc. in Computer Science and Engineering · GPA: 5.0/5.0

May 2024

Relevant coursework: software construction, advanced algorithms, distributed systems, operating systems, compilers, formal verification, FPGA programming, security, and mathematics (linear algebra and probability).

WORK EXPERIENCE

Citadel · Global Quantitative Strategies

New York, NY

Quantitative Research Engineer

Jul 2025–now

- Built and maintained components of a firmwide live and historical market data platform for research and trading.
- Developed automation and developer tooling to improve system reliability, observability, and development speed.
- Designed and implemented streamlined workflows for onboarding new market datasets, improving data quality and operational robustness and enabling rapid collaboration with alpha researchers.

Jump Trading · Jump Core Strategies

Chicago, IL

Software Engineer Intern

Jun 2024–Aug 2024

- Developed an auto-vectorizing just-in-time (JIT) compiler for an internal domain-specific language (DSL) in C++.
- Enabled efficient execution of latency-critical workloads by leveraging SIMD vectorization and runtime specialization.

Instabase

Menlo Park, CA

Software Engineer Intern

May 2023–Aug 2023

- Designed and implemented distributed job orchestration framework in Go using Kubernetes and Jenkins to manage complex deployment workflows.
- Led migration from legacy system to new framework; improved job success rates through better observability, and built-in fault tolerance.

DRW Holdings

Chicago, IL

Software Developer Intern

Jun 2022–Aug 2022

- Extended internal data pipelines to integrate additional pricing sources for downstream trading and research systems.
- Profiled and optimized pipeline services, eliminating compute and memory bottlenecks of internal tools.

Frictionless Systems

Cambridge, MA

Software Engineer Intern

Feb 2021–May 2021

RESEARCH AND TEACHING EXPERIENCE

MIT CSAIL–Programming Languages and Verification Group

Cambridge, MA

Undergraduate and graduate research assistant

Jan 2023–May 2025

MIT–Department of Electrical Engineering and Computer Science

Cambridge, MA

Teaching assistant — C and assembly programming · Computer Architecture · Compilers

Sep 2021–May 2025

AWARDS AND HONORS

Frederick C. Hennie III Teaching Award · Outstanding contribution to departmental teaching

May 2025

Northern Telecom/BNR Project Award · Best student project in Digital Systems Laboratory (6.111)

May 2025

International Olympiad in Informatics · Bronze Medal

Sep 2018

Asia–Pacific Informatics Olympiad · Silver Medal

May 2018

Thailand Olympiad in Informatics · Gold Medal

June 2017

SKILLS

Languages: Python, C, C++, Rust, Go, Javascript, Typescript, Java, Scala, Kotlin, Haskell, C#, SystemVerilog